Predicting Social Media Engagement Through Machine Learning: Evidence, Methods and Implications for Communications Practice

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Abstract

Social media analytics have traditionally allowed communications professionals to evaluate content only after it has been published. Reach, impressions, likes, comments, shares, saves, clicks and watch time tell an organisation what happened. A harder and more useful question is whether the historical record of past posts can be used to estimate, with known and acceptable error, how a new post is likely to perform before it goes out.

This paper examines what the evidence says about predicting social media engagement with statistical and machine-learning methods, and what that evidence means for everyday communications work. It draws on peer-reviewed engagement-prediction studies, large industry benchmark datasets covering tens of millions of posts, global platform-use statistics for 2025 and 2026, and recent work on hybrid ensemble models, context-aware deep learning and Explainable Artificial Intelligence (XAI).

The evidence shows that engagement is predictable to a useful but limited degree. Models trained on an organisation's own Facebook or Instagram history can forecast post-level engagement better than simple averages, and richer models that read text, images and context perform better still. However, prediction quality depends heavily on how the outcome is defined, whether the model uses only information genuinely available before publication, how it is tested, and whether it is compared against a simple baseline. Historical correlations are not causal rules, and an accurate engagement forecast is not the same thing as effective communication.

The paper explains these ideas for readers without a technical background, sets out a step-by-step research design a communications team can apply to its own data, and argues that predictive modelling is most valuable as decision support that sharpens human editorial judgement rather than replacing it.

Throughout, the paper uses the image of a leaking cup. Communication leaks in two ways: as lost impact, when effort never reaches people, and as lost knowledge, when lessons are never captured. Predictive modelling, used well, is a way of finding those leaks before

the next pouring rather than measuring the puddle afterwards.

Keywords: social media engagement; predictive analytics; machine learning; regression; natural language processing; explainable AI; SHAP; communications strategy; data leakage; decision support

1. Introduction

The leaking cup

Picture a cup. Into it, a communications team pours everything it has: staff time, carefully made content, a modest budget, and above all the stories, words and images of the people and communities it works with. The cup is meant to carry that effort to an audience, so that something real happens: understanding grows, trust deepens, people apply, attend, act.

But every cup leaks, in two ways. The first is a leak of **impact**: effort that never reaches people. Posts are buried by feeds, audiences are missed, and reactions are counted without anything changing. The second is a leak of **knowledge**: lessons the team never captures. Results go unrecorded, assumptions go untested, and what was learned last year drains away before next year's plans are made.

Most social media analytics measure the puddle on the table after the pouring is done. They tell us, accurately and in great detail, how much was spilled. They rarely tell us where the holes are. This paper is about a different ambition: using what the puddle has taught us to find the leaks *before* we pour again. Prediction cannot give us a new cup, and it will never stop every drop. But it can help a team see where its effort is most likely to escape, and it can turn each pouring into a record that makes the next one wiser.

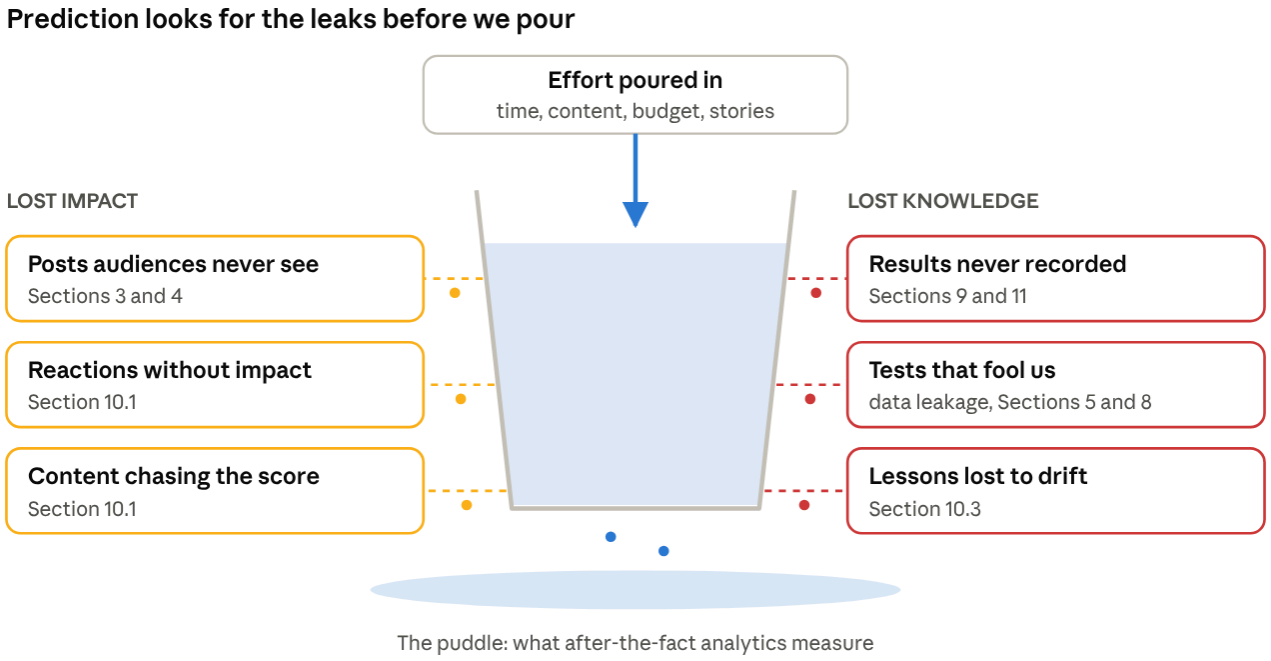
A cup designed to leak

No one knows who made the first cup; drinking vessels are older than written history. But one cup is famously credited to a named inventor, and it was built to leak on purpose. The Pythagorean cup, also called the greedy cup or the cup of justice, is traditionally attributed to Pythagoras of Samos, the Greek philosopher and mathematician of the sixth century BCE best remembered for the theorem that bears his name.

The cup looks ordinary except for a column rising from its centre. Hidden inside is a bent tube that works as a siphon. Fill the cup to a modest line and it behaves like any other cup. Fill it past that line and the liquid spills into the tube and drains out through the base, until the cup is completely empty. According to legend, Pythagoras designed it to teach his

students moderation and to make sure everyone shared fairly. Physicists who use the cup in teaching note that it is unclear whether Pythagoras actually invented it, though the earliest examples are thought to date from roughly his era.

Whether or not the attribution is true, the lesson fits this paper closely. The Pythagorean cup does not punish pouring; it punishes pouring without regard for the line. A communications team that fills its feeds faster and faster in pursuit of reactions can find, like the greedy drinker, that the very act of chasing more empties the cup of what mattered. Knowing where the line is, and where the leaks are, is the kind of knowledge this paper is about.



the leaking cup · where impact and knowledge escape, and the sections that address each

The image returns throughout the paper. Each section addresses one of the leaks shown above, and the closing section asks what a team can realistically do to hold more of what it pours.

Social media is now part of the basic infrastructure of public communication. According to the Digital 2026 Global Overview Report, there were about 5.66 billion active social media user identities worldwide in October 2025, equal to roughly 68.7 per cent of the global population, after growth of 4.8 per cent (259 million new identities) in twelve months. These are counts of "user identities" rather than unique people, because one person may hold several accounts, but the order of magnitude is not in doubt.

For communications professionals, this scale creates a measurement problem as much as an opportunity. Every published post generates data: reach, impressions, reactions, comments, shares, saves, link clicks, video views and follower changes. Most dashboards

use this data to answer retrospective questions. Which post performed best last month? Which platform delivered the most reach? Did video outperform still images? These questions are useful, but they all look backwards.

Predictive analytics asks a different question: can information that is known *before* a post is published be used to estimate how that post is likely to perform? The difference matters. Descriptive analytics finds patterns in what has already happened. Predictive analytics tests whether those patterns are stable enough to say something reliable about what has not happened yet. A pattern that merely describes the past may or may not survive that test.

This paper is organised around one central research question:

> To what extent can regression and machine-learning techniques use historical social media data to predict the engagement of future posts, and how can such predictions be used responsibly in communications practice?

The paper has four objectives. First, it explains the core ideas of predictive modelling in plain but precise language, so that readers without a background in artificial intelligence (AI) can follow the argument. Second, it reviews the empirical evidence on whether, and how well, social media engagement can be predicted. Third, it sets out a research design that a communications team could apply to its own data. Fourth, it examines the limits of engagement prediction, including the gap between engagement and genuine communications impact, the problem of mistaking correlation for causation, and the ethical and contextual questions that arise when organisations, particularly in African and civil-society settings, begin to use these tools.

A note on the numbers

Scale figures like these are useful context, but they need careful reading. DataReportal itself cautions that its counts of user identities "may not represent unique human individuals." — DataReportal, Digital 2026 Global Overview Report

That caution sets the tone for the rest of this paper. Social media numbers are measurements produced by platforms, with their own definitions and blind spots. A predictive model built on those numbers inherits every one of those blind spots, so understanding where the data come from is the first step in any serious prediction work.

Why the question matters now

Three developments make the question of prediction more pressing than it was a decade ago. First, the volume of content has grown faster than audiences' attention, so most organisations now compete for visibility in crowded feeds. Second, platform algorithms increasingly decide who sees a post, which makes performance less transparent and

harder to explain by intuition alone. Third, the tools needed to build simple predictive models, such as spreadsheet software, free Python notebooks and open-source libraries, are now available to small teams at little or no cost. The question is no longer whether a communications team *can* build a model, but whether it can build one that is honest, useful and used wisely.

An everyday example

Consider a communications officer planning next week's posts. They have five items: a grant announcement, a partner story, a photo album from a workshop, a short video interview and a policy commentary. Experience suggests the grant announcement will do well and the policy commentary less so. A predictive model does not replace that experience. It tests it: are grant announcements really stronger on average once format, timing and audience size are accounted for? Does the workshop album perform better as a carousel than as a single image? Is Thursday genuinely a better day than Monday for this audience, or does it only seem so because important posts happened to go out on Thursdays? The rest of this paper explains how such questions can be asked and answered with evidence.

Questions to consider

1. When your team reviews social media performance today, are you mainly describing what happened, or trying to anticipate what will happen?
2. Which of your current beliefs about "what works" have been tested against data, and which rest mainly on memory or impression?
3. What decisions would you make differently if you had a reasonably reliable forecast of how a post was likely to perform?
4. What would be lost if decisions were made *only* on the basis of such forecasts?

2. Key concepts for non-specialist readers

A small number of ideas carry most of the weight in this paper. They are explained here once, so that later sections can use them without repeated definitions.

2.1 Engagement

"Engagement" is the set of visible actions people take in response to a post: reactions or likes, comments, shares or reposts, saves, and sometimes clicks or video views. It is often summarised as a total (likes + comments + shares + saves) or as an engagement *rate*, which divides that total by the number of followers or the number of people reached. A systematic literature review by Trunfio and Rossi (2021) confirms that engagement is a

multidimensional and loosely defined concept, and that platforms, agencies and scholars measure it in different ways. Any prediction project must therefore state precisely which version of engagement it is predicting.

2.2 Prediction and forecasting

To predict, in this paper, means to produce an estimate of a quantity that is not yet known, using information that is already known. A weather forecast predicts tomorrow's rainfall from today's measurements. An engagement model predicts how many engagements a planned post will receive from things like its format, topic, planned posting time and the account's audience size. A prediction is always an estimate with error attached; the question is never "is the prediction exactly right?" but "how wrong is it, on average, and is that good enough to be useful?"

2.3 Variables: outcome and predictors

Every predictive model has one **outcome variable** (also called the target), which is the thing being predicted, such as "engagements within 48 hours". It also has a set of **predictor variables** (also called features or inputs), which are the pieces of information the model uses to make its estimate, such as content format or posting hour. The quality of a model depends at least as much on how carefully these variables are chosen and measured as on the sophistication of the mathematics.

2.4 Statistical models and machine learning

A **statistical model** is a mathematical rule that links predictors to an outcome. The simplest and most widely used is **linear regression**, which assumes that each predictor adds or subtracts a fixed amount from the predicted outcome. **Machine learning** refers to a family of methods in which the computer learns the rule from examples rather than having it written by hand. Some machine-learning methods, such as decision-tree ensembles and neural networks, can capture patterns that are curved, conditional or interacting (for example, "video performs well, but only on weekends and only for event content"), which simple linear regression cannot represent without help. In practice the boundary between "statistics" and "machine learning" is blurred; linear regression is itself often taught as the most basic machine-learning method.

2.5 Training and testing

A model learns from **training data**: a set of past posts for which both the predictors and the actual outcome are known. Its quality is then judged on **test data**: a separate set of past posts that the model was not allowed to see during learning. This separation is the single most important safeguard in predictive work. A model that is graded on the same posts it learned from is like a student who is examined on the exact questions they

memorised; the score says little about how they will cope with new questions. When the goal is to predict the future, the most realistic approach is to train on older posts and test on newer ones.

2.6 Baseline

A **baseline** is the simplest reasonable way of making the same prediction without a model, for example "assume every new post will receive the account's historical average engagement". A model is only worth using if it consistently beats this baseline on test data. This idea sounds modest, but it is frequently skipped, and skipping it is one of the main reasons organisations overestimate the value of complex tools.

2.7 Overfitting and generalisation

Overfitting happens when a model learns the noise and accidents of its training data rather than the underlying pattern. An overfitted model looks excellent on training data and disappointing on new data. **Generalisation** is the opposite: the ability to perform well on data the model has not seen. More flexible models are more powerful but also more prone to overfitting, especially when the dataset is small, which is the normal situation for a single organisation's social media accounts.

2.8 Correlation and causation

A predictive model exploits **correlations**, meaning patterns in which two things tend to occur together. It does not, by itself, establish **causation**, meaning that changing one thing will produce a change in the other. If posts published on Wednesdays have historically received more engagement, a model may correctly use "Wednesday" to improve its forecasts. That does not show that moving a post to Wednesday will raise its engagement; perhaps the organisation simply tended to publish its most important announcements on Wednesdays. This distinction recurs throughout the paper.

2.9 Why definitions matter so much

> Their review confirms "the polysemic and multidimensional nature of social media engagement." — Trunfio and Rossi (2021)

"Polysemic" means having many meanings. In practice, it means two teams can both say "our engagement went up" while measuring completely different things: one counting reactions, another counting comments, a third dividing by followers and a fourth by reach. When a model is built, that ambiguity cannot survive. The computer needs one number per post, calculated the same way every time. Settling the definition is therefore not a technicality but the first real research decision.

2.10 Quick reference

Term	Plain meaning	Example in communications
Outcome (target)	The number you are trying to predict	Engagements within 48 hours
Predictor (feature)	A piece of information the model uses	Format, topic, posting hour
Training data	Past posts the model learns from	Posts from 2024 and 2025
Test data	Past posts held back to check the model	Posts from early 2026
Baseline	The simplest sensible forecast	"Same as our usual median"
Overfitting	Learning noise instead of pattern	A model that "remembers" one viral post
Correlation	Things that tend to occur together	Videos and high engagement
Causation	One thing producing another	Switching to video *raises* engagement

2.11 Questions to consider

1. If you asked three colleagues to define "engagement" for your organisation, would they give the same answer?
2. Which of your current reports mix outcomes that should be kept separate, such as reach and engagement?
3. Can you think of a pattern in your own data that might be a correlation rather than a cause?
4. How many past posts do you actually have for each platform, and is that enough to hold some back for testing?

3. The contemporary social media environment, 2024–2026

Large industry datasets do not prove that engagement can be predicted, but they show that engagement is not randomly distributed. It varies systematically with platform, content format, account behaviour and audience size. Wherever there is systematic variation, there is at least the possibility of prediction.

3.1 Scale and platform mix

Global adoption has been summarised above. Platform mix differs sharply by country and region, which matters because a model trained on one platform's data rarely transfers cleanly to another. In the United States, Pew Research Center's 2025 survey of 5,022 adults found that 84 per cent use YouTube, 71 per cent use Facebook and about half use Instagram, with smaller shares using TikTok (37 per cent) and WhatsApp (32 per cent). Across eight countries Pew surveyed in Latin America, Africa and South Asia, by contrast, a median of 73 per cent of adults reported using WhatsApp and 62 per cent Facebook.

Country-level data also come with important caveats. DataReportal's Digital 2026 report on Uganda counted 2.40 million social media user identities in October 2025, equal to 4.6 per cent of the population, while cautioning that its method may sometimes count only the users of the single largest platform in a country. In Uganda's case, Facebook has been officially blocked since 2021, so platform-reported audiences are likely to understate real use that happens through VPNs. For an organisation working in East Africa, this is a reminder that audience-size variables drawn from platform tools may be noisy or biased before any modelling begins.

3.2 Format differences on Instagram

Socialinsider's 2026 Instagram benchmarks illustrate how much average performance differs by format. On its dataset, carousels reached an average engagement rate of about 0.55 per cent and Reels about 0.52 per cent, with single images lower. Its quarterly figures for 2026 show the same ordering, together with a general decline across all formats:

Instagram format	Engagement rate, Q1 2026	Engagement rate, Q2 2026
Carousels	0.52%	0.50%
Reels	0.50%	0.48%
Images	0.35%	0.33%

Average Instagram engagement rate by format, as reported in Socialinsider's quarterly benchmarks.

It is tempting to read this table as a rule: "publish more carousels." That conclusion would be premature. An average across hundreds of thousands of accounts says nothing definite about whether *one particular* organisation's next carousel will outperform its next Reel. Account size, subject matter, audience composition, visual quality and timing all intervene. The useful lesson is narrower and more important: content format carries information that may help a model, and so it is a candidate predictor.

3.3 Evidence from more than 52 million posts

Buffer's *State of Social Media Engagement 2026* analysed more than 52 million posts from over 200,000 accounts across platforms including Instagram, TikTok, LinkedIn, X, Threads, Facebook and Bluesky, published between January 2024 and December 2025. It defined engagement rate as likes, comments, shares and saves divided by followers. One of its most consistent findings was behavioural rather than about format: across nearly two million posts on six platforms, accounts that replied to comments outperformed those that did not, by as much as 42 per cent on Threads and 30 per cent on LinkedIn.

This finding illustrates both the promise and the trap of large descriptive datasets. It suggests that account behaviour, not only content, contains predictive signal. But it does not prove that replying *causes* higher engagement; accounts that reply may also be more active, better resourced or already more popular. Buffer itself presents its medians as typical of its own user base rather than universal benchmarks.

3.4 Candidate predictors suggested by the evidence

Taken together, the descriptive evidence suggests a set of variables that an organisation could test as predictors: platform; content format; topic or campaign category; day and hour of publication; caption length; number of hashtags and mentions; whether the post is paid or organic; video duration; audience size at the time of posting; the account's recent average performance; whether the team typically responds to comments; and characteristics of the text and images. The research question is not whether these variables differ on average, which they clearly do, but which of them continue to improve forecasts for posts the model has never seen.

The leaking cup. Before looking for leaks, it helps to admit that we cannot see the whole cup. Platform figures count identities rather than people, restrictions push audiences behind VPNs, and benchmarks describe other organisations' cups, not ours. Some of the water we think is missing was never measured in the first place. Any model inherits these blind spots, so the first discipline is to know the shape of our own cup before comparing it with anyone else's.

3.5 How benchmark publishers describe their own data

The organisations that publish large benchmarks are often careful about what their numbers mean, and their own words are worth noting.

> Buffer describes its medians as representing "typical performance in Buffer's dataset, rather than universal benchmarks." — Buffer, State of Social Media Engagement 2026

In other words, the figures describe the accounts that use Buffer's tools. Those accounts may differ systematically from a Ugandan civil-society organisation, a government ministry or a university in Nairobi.

> Socialinsider concludes that "Carousels remain Instagram's highest-performing content format." — Socialinsider, 2026 Instagram Organic Benchmarks

This is a statement about averages across its dataset. Whether it holds for any one account is exactly the kind of question a predictive model, tested on that account's own history, can answer.

3.6 Reading a benchmark: a worked example

Suppose your organisation's Instagram account has a median engagement rate of 1.2 per cent, well above the platform averages in the table in Section 3.2. It would be wrong to conclude that the account is "beating the market" in any meaningful sense, or that copying the benchmark's format mix would help. Smaller accounts commonly have higher engagement rates than very large ones, because a larger share of their followers are personally connected to the organisation. The right comparison is the account against its own history, and the right question is which of its own posts outperformed its own typical level, and why. Benchmarks give context; an organisation's own data give evidence.

3.7 Questions to consider

1. Which external benchmarks does your team currently quote, and do you know whose accounts they are based on?
2. How similar is your audience, in size, location and interests, to the accounts behind those benchmarks?
3. Which platforms really matter for your audience, and are the platforms you report on the same ones?
4. Are there gaps in your audience data, for example because of platform restrictions or shared devices, that a model would not see?

4. What the research literature shows

The academic evidence supports a cautious but positive conclusion: post-level engagement can be predicted better than chance and better than simple averages, typically with average errors in the range of 15 to 30 per cent of the true value, and the best results come from models that combine several kinds of information. The evidence also shows clear limits.

4.1 Foundational evidence: predicting from pre-publication features

One of the most widely cited studies is Moro, Rita and Vala (2016), published in the *Journal of Business Research*. The authors modelled twelve performance metrics for 790 posts from a cosmetics brand's Facebook page, using only seven inputs that were known before publication: content category, page total likes, post type, month, hour, weekday and whether the post was paid. Their two best models achieved a mean absolute percentage error of around 27 per cent, and the type of content was the single most influential input, with a relative importance of about 36 per cent. Part of this dataset is publicly available through the UCI Machine Learning Repository, which explicitly separates the seven pre-publication features from the twelve post-publication outcome measures. That separation is a model of good practice, and Section 5 returns to why it matters.

The study is now a decade old, and platforms have changed a great deal since 2014. Its lasting value is methodological: it shows that a modest dataset from a single organisation, using only information available at the moment of planning, can support useful forecasts.

4.2 Recent evidence: multiple platforms, text and explainability

Vairetti, Gubbins and Maldonado, writing in *Social Network Analysis and Mining*, built a media-analytics framework for a major Chilean television channel's Facebook and Instagram posts. They combined traditional statistical modelling, natural language processing and explainable machine learning. On Facebook, Extreme Gradient Boosting (XGBoost) performed best, with a mean absolute percentage error of 25.14 per cent; on Instagram, Support Vector Regression performed best, with an error of 16.56 per cent. Two points stand out. First, the error levels are similar to those reported by Moro and colleagues, which suggests that roughly 15 to 30 per cent average error is a realistic expectation for organisational accounts. Second, the best algorithm differed by platform, which means there is no single "best model" that a communications team can adopt without testing.

In the higher-education sector, Bhattacharjee, Suresh and Maiti (2026) built predictive models of social media engagement for higher-education institutions in an emerging-market context, selecting the best models through k-fold cross-validation, and argue that institutions which post without a strategy do not gain awareness or enrolments simply by being present.

4.3 Evidence from user-level behaviour at very large scale

A different strand of research predicts the behaviour of users rather than the performance of posts. Peters and colleagues (2024), working with Snap Inc. and Columbia University, applied deep LSTM neural networks to more than 100 million Snapchat

sessions from almost 80,000 users. Past behaviour alone explained about a third of the variation in active versus passive use ($R^2 = 0.345$); adding context information such as connectivity status, location, time and weather raised this to just over half ($R^2 = 0.522$). With minimal behavioural history plus momentary context, the model still explained about 44 per cent of the variation. This work is discussed in more detail in Section 7. Its main relevance here is that context, not only content and history, carries predictive information, and that richer context can partly replace long, privacy-intensive behavioural histories.

4.4 Summary of selected studies

Study	Data	Outcome	Best reported result
Moro, Rita & Vala, 2016	790 Facebook posts, one cosmetics brand	12 post metrics	MAPE ≈ 27%
Vairetti, Gubbins & Maldonado, 2026	Facebook and Instagram posts, Chilean TV channel	Post engagement	MAPE 25.14% (Facebook, XGBoost); 16.56% (Instagram, SVR)
Peters et al., 2024	100M+ Snapchat sessions, ~80,000 users	Active vs passive use	R^2 0.522 with context (0.345 without)
Bhattacharjee, Suresh & Maiti, 2026	Higher-education institutions, emerging market	Engagement class	Best model chosen by k-fold cross-validation

MAPE = mean absolute percentage error (lower is better). R^2 = share of variation explained (higher is better). These figures come from different datasets and outcomes and cannot be compared as a league table.

4.5 The limits of predictability

There is also strong evidence that some social outcomes are inherently hard to predict. In a well-known online experiment, Salganik, Dodds and Watts (2006) showed that when people could see what others had downloaded, the success of songs became both more unequal and less predictable: the same song could succeed in one "world" and fail in another, purely through early chance and social influence. Martin and colleagues (2016) later estimated that even with complete information about a post and its author, the predictability of how widely content spreads on Twitter is bounded well below perfect.

And in a large scientific mass collaboration, Salganik and colleagues (2020) found that 160 teams using sophisticated machine-learning methods could predict life outcomes only slightly better than a simple baseline model.

For communications teams, these studies suggest a realistic stance. Ordinary, recurring content (announcements, event updates, programme stories) is likely to be moderately predictable. Rare viral successes are, almost by definition, hard to predict, because they depend on chance early sharing and on the actions of a few highly connected people. A model is a tool for improving average planning decisions, not for finding guaranteed hits.



predictability spectrum · from routine updates to viral breakouts

The further right a post sits on this spectrum, the more its success depends on chance and social influence rather than on features a model can see.

4.6 In the researchers' own words

The studies above are worth hearing directly, because their authors chose their words carefully.

> Their Facebook and Instagram results "confirm the predictive capabilities of machine learning." — Vairetti, Gubbins and Maldonado

This is the optimistic case: on real organisational data, prediction works well enough to be useful.

> "Experts routinely fail to predict which products will succeed." — Salganik, Dodds and Watts (2006)

This is the cautionary case. In their music-market experiment with 14,341 participants, the authors also found that the best songs rarely did poorly and the worst rarely did well, but almost any other outcome was possible. Quality mattered, but it did not settle the result.

> On Twitter, their "best performing models can explain less than half of the variance in cascade sizes." — Martin, Hofman, Sharma, Anderson and Watts (2016)

This held despite what the authors describe as an unprecedented volume of information about users, content and past performance. Their conclusion is that even unlimited data would not make the spread of content fully predictable.

Read together, these voices are not contradictory. They describe two regions of the same landscape. Ordinary, recurring content can be forecast with useful accuracy. Exceptional, socially amplified success cannot be reliably foreseen, however good the model.

The leaking cup. Some water will always escape in ways no one can foresee. A post that is shared by the right person at the right moment splashes far beyond the cup; an equally good post sinks without trace. Prediction is not a promise to catch every drop. Its realistic aim is to stop the steady, repeated leaks in routine communication, which over a year carry away far more effort than any single viral moment brings in.

4.7 Questions to consider

1. Which of your posts in the last year would you place at each point on the predictability spectrum?
2. Are your planning decisions mostly about routine content, where prediction helps, or about rare breakthroughs, where it helps much less?
3. When a post "goes viral", does your team treat it as a repeatable formula or as partly good fortune?
4. Which of the studies above is closest to your own organisation in size, platform and audience, and what does that suggest about the accuracy you could expect?

5. Defining a prediction problem

The first and most consequential decision in any prediction project is not which algorithm to use but exactly what is being predicted, from what information, and at what moment.

5.1 Choosing the outcome

"Engagement" can mean several different things, and each choice answers a different communications question. A common composite is:

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\text{Total engagement} = \text{Likes} + \text{Comments} + \text{Shares} +
\text{Saves}
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But an organisation might instead care most about comments (conversation), shares (onward dissemination), link clicks (driving people to a report or application form), watch time or completion rate (for video), or website visits. Each of these can be modelled, and they will often have different predictors.

The outcome also needs a fixed time window. "Engagements within 48 hours of publication" is far easier to compare across posts than "lifetime engagements", because older posts have simply had longer to accumulate reactions. Without a fixed window, a model may learn that older posts perform better for no reason other than their age.

Finally, the team must choose between predicting a **count** (for example, 126 engagements), a **rate** (engagements divided by followers or reach), or a **category** (for example, "below normal", "normal" or "above normal" for this account). Counts are intuitive but strongly influenced by audience size. Rates allow fairer comparison across accounts. Categories are often more robust and easier to act on when data are limited.

5.2 Choosing predictors and avoiding data leakage

Predictors should be chosen because there is a plausible reason they might relate to engagement and because they can be measured consistently. Equally important is a rule that is easy to state and easy to break:

> A model may only use information that would genuinely be available at the moment the prediction is made.

Breaking this rule is called **data leakage**. Suppose the aim is to predict engagement before a post goes live, but the model is given the post's final reach or lifetime impressions. Those numbers are only known after publication and are themselves closely tied to engagement. The model will appear extraordinarily accurate in testing and be useless in real planning, because those inputs will never be available when a real decision is being made. The scikit-learn project, which maintains one of the most widely used machine-learning libraries, lists data leakage among the most common pitfalls in practice and recommends that all data preparation steps be learned from training data only.

Leakage can also be subtle. "Average engagement of this account" is a legitimate predictor only if it is calculated from posts published *before* the post being predicted. If it is calculated over the whole dataset, including future posts, the model is quietly being shown the answers.

A practical test for every candidate predictor is: *Would I know this value at the exact moment I want the model to make its prediction?* If not, it does not belong in a pre-publication model.

The leaking cup. Here the metaphor becomes literal. Data leakage is a hidden crack that runs the wrong way: instead of water escaping, information from the future seeps in. In testing, the cup looks full to the brim. In real use, the moment the future is no longer available, it empties. This is a leak of knowledge of the most dangerous kind, because it produces false confidence rather than an honest gap.

5.3 The regression model

Regression is the natural starting point when the outcome is a number. In its simplest form, with a single predictor X:

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\hat{Y} = \beta_0 + \beta_1 X
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Here $\hat{Y}$ (read "Y-hat") is the predicted engagement, β_0 is the intercept (the predicted value when X is zero), and β_1 is the slope (how much the prediction changes when X increases by one unit). With several predictors, the model becomes:

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Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon
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where ε (epsilon) represents everything the model does not capture. For example, X_1 might be audience size before publication, X_2 the content format and X_3 the posting hour. Categorical information such as format (image, carousel, Reel, video) is converted into a set of yes/no indicators before the model is fitted, so that each format receives its own estimated effect relative to a reference format.

A worked illustration. Assume a simplified model in which predicted engagement depends only on audience size, measured in units of 10,000 followers:

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\hat{Y} = 20 + 0.6X
```

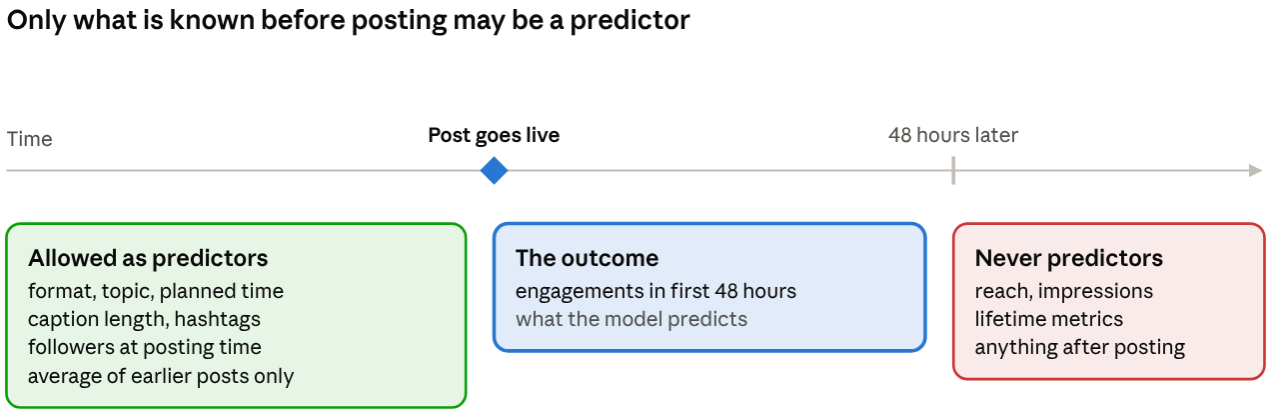
For an account with 500,000 followers, X = 500,000 ÷ 10,000 = 50, so the predicted engagement is 20 + (0.6 × 50) = 50 engagements. The example shows only the mechanics. It does not imply that follower count *causes* engagement, nor that the relationship is a straight line in real data; in practice, engagement *rates* usually fall as accounts grow larger.

Regression has real strengths for communications teams. Its coefficients can be read directly ("holding other factors constant, a carousel is associated with about N more engagements than a single image"), it needs relatively little data, and it is less prone to overfitting than flexible machine-learning methods. Its main weakness is that, unless the analyst deliberately adds them, it cannot represent curved relationships or interactions between variables.

5.4 Practical transformations

Social media engagement data are usually highly skewed: most posts receive modest engagement and a few receive very large numbers. Modelling the logarithm of engagement rather than the raw count often produces more stable models and stops a handful of viral posts from dominating the estimates. Audience size is similarly often better entered as a logarithm. These are standard choices, but they should be documented, because they change how coefficients and errors are interpreted.

5.5 The leakage rule at a glance



the leakage rule on a time line · allowed predictors, outcome window, forbidden inputs

Everything to the left of the moment a post goes live can be used to predict it; the outcome is measured in the window after; and nothing measured after publication may ever be fed back in as a predictor.

The public version of the Moro et al. dataset shows how this separation looks in practice. Its documentation states that it includes "7 features known prior to post publication" alongside separate measures of impact. — UCI Machine Learning Repository, Facebook Metrics

A beginner who downloads that dataset and uses all 18 columns to predict engagement will get a strikingly accurate model, and it will be useless for planning, because most of those columns describe what happened after the post was published.

5.6 A short self-test

For each candidate predictor below, decide whether it belongs in a model that predicts engagement *before* publication. Suggested answers follow in brackets.

1. The planned posting day and hour. (Yes: decided in advance.)
2. The number of people who saw the post in its first hour. (No: only known after publication; it belongs in a post-publication model.)

3. Follower count on the morning of publication. (Yes: known at the time.)
4. Whether the post was later boosted with paid promotion. (Only if the decision to boost is made before publishing; if boosting depends on early performance, it is leakage.)
5. The account's average engagement over all posts in the dataset. (No, if the average includes posts published after this one; yes, if it is computed only from earlier posts.)
6. The topic category assigned by the communications team when planning. (Yes.)

5.7 Questions to consider

1. Which outcome would best reflect what your communication is for: total engagement, comments, shares, clicks or something else?
2. Does your team record posting plans (format, topic, time) in a way that could later be matched with results?
3. Are any of your current "insights" based on metrics that would only be known after a post is live?
4. Would a count, a rate or a simple category (below, around or above normal) be most useful for your planning decisions?

6. Reading the meaning of content: language, sentiment and images

Structured variables such as format, day and audience size capture the packaging of a post. They say nothing about what the post is actually *about*. Two carousels published at the same hour by the same account can perform very differently because one announces a grant opportunity and the other reports on an internal meeting. Natural language processing (NLP) and computer vision are the techniques that allow some of this meaning to be measured.

6.1 Turning words into measurable features

The simplest text features need no specialised AI at all: caption length, number of hashtags and mentions, presence of a question, presence of a call to action ("apply now", "read more"), presence of a link, readability scores and the use of emojis. These are easy to compute in a spreadsheet and often carry real signal.

More advanced approaches represent the *meaning* of text. **Topic models** such as Latent Dirichlet Allocation (Blei, Ng and Jordan, 2003) scan a large collection of posts and identify recurring themes, for example "funding opportunities", "partner stories" or "policy commentary", and estimate how much of each theme a post contains. **Language models** such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) convert a caption

into a long list of numbers, called an *embedding*, in which captions with similar meaning sit close together even when they use different words. These embeddings can then be fed into a predictive model as features.

The practical gain is that the organisation can move beyond asking "does video beat images?" to asking "which combinations of topic, platform and format tend to do better for our audience?" That is a more useful planning question.

6.2 Sentiment analysis and its limits

Sentiment analysis classifies text as positive, negative or neutral, or gives it a score on a scale. It is widely used in brand monitoring and can be added as a predictor. It must be interpreted with care. A post about an urgent injustice, a funding cut or a humanitarian crisis may be "negative" in tone and yet generate exceptionally strong, constructive engagement because the issue matters to the audience. Positive sentiment is not the same as good communication, and a model that learns "positive posts do better" from a particular historical period could push a mission-driven organisation towards blandness.

Sentiment tools also perform unevenly across languages and registers. Most widely used models were trained mainly on English and other high-resource languages. Content in Luganda, Swahili or other African languages, or captions that mix English with local languages, may be scored poorly or not at all. Multilingual models developed specifically for African languages, such as AfriBERTa (Ogueji, Zhu and Lin, 2021), are a partial response, but any organisation working in multilingual settings should test text features carefully on its own content before trusting them.

6.3 Images and video

Social media is increasingly visual. Computer-vision models, typically deep convolutional networks such as ResNet (He et al., 2016) that were pre-trained on millions of labelled images, can extract features such as whether people are present, how many faces appear, the dominant colours, image brightness and quality, how much text is overlaid on the image, and whether the setting is indoors or outdoors. For video, additional features include duration, the number of scene changes, and, after publication, retention and completion curves.

A mature prediction system may combine three kinds of information about each post: structured metadata (format, timing, audience), text features (topic, tone, embeddings) and visual features. This is usually called **multimodal** modelling. It is more powerful but also more demanding: it needs more data, more computing, and more care to avoid overfitting. For most communications teams, structured metadata plus a small set of simple text features is the right place to start.

6.4 What the research says about combining signals

Vairetti, Gubbins and Maldonado took this combined approach in their study of a Chilean television channel's posts. They joined traditional statistical modelling of post characteristics with natural language processing of the text, and then used explainable machine learning to see which features mattered most.

The reason to combine signals is simple. A strong image with a weak caption, or a compelling caption posted when the audience is offline, may underperform. A model that sees several kinds of information at once (what a post looks like, what it says, and when it reaches people) has more to work with than one that sees only a single kind, provided there is enough data to support it.

6.5 Worked example: from one post to a row of numbers

Imagine this hypothetical Instagram caption from a civil-society organisation:

> *Applications are now open for our 2027 Community Media Fellowship! Are you a young storyteller in Uganda? Apply by 30 November — link in bio. #Fellowship #UgandaYouth*

Accompanied by a photograph of three smiling past fellows outdoors, it could be turned into features like these:

Feature	Value	How it was obtained
Format	Single image	Recorded when planning
Caption length	About 150 characters	Counted automatically
Hashtag count	2	Counted automatically
Contains a question	Yes	Simple text rule
Call to action	Yes ("Apply")	Simple text rule
Topic	Opportunity / call for applications	Team label or topic model
Sentiment	Positive	Sentiment tool (checked by a person)
People in image	Yes, three faces	Computer-vision model
Setting	Outdoors	Computer-vision model

None of these values, taken alone, tells us whether the post will do well. Together, and compared with hundreds of earlier posts described in the same way, they allow a model to estimate how this post is likely to perform relative to the account's usual level.

6.6 Questions to consider

1. In which languages does your organisation publish, and have any text tools you use been tested on those languages?
2. Do your most important posts tend to be positive in tone, or do they often deal with difficult issues? What would a sentiment-based model learn from that?
3. Which simple text features (questions, calls to action, length) could you start recording today without any AI?
4. What visual choices does your team make deliberately, and could they be recorded as features?

7. Hybrid machine learning, context-aware models and explainable AI

Linear regression is the right starting point, but social media behaviour is rarely a set of straight-line relationships. The effect of posting time may depend on the platform. The value of a Reel may depend on the topic. An extra 10,000 followers matters more to a small account than to a very large one. Machine-learning methods can capture these curved and conditional patterns, and explainability tools can then show communications teams what those models have learned.

7.1 Decision trees and ensembles

A **decision tree** makes a prediction by asking a sequence of yes/no questions: "Is the format video? If yes, was it posted on a weekend? If yes, is the topic an event?" Each path through the tree ends in a predicted value. Single trees are easy to read but unstable: small changes in the data can produce a very different tree.

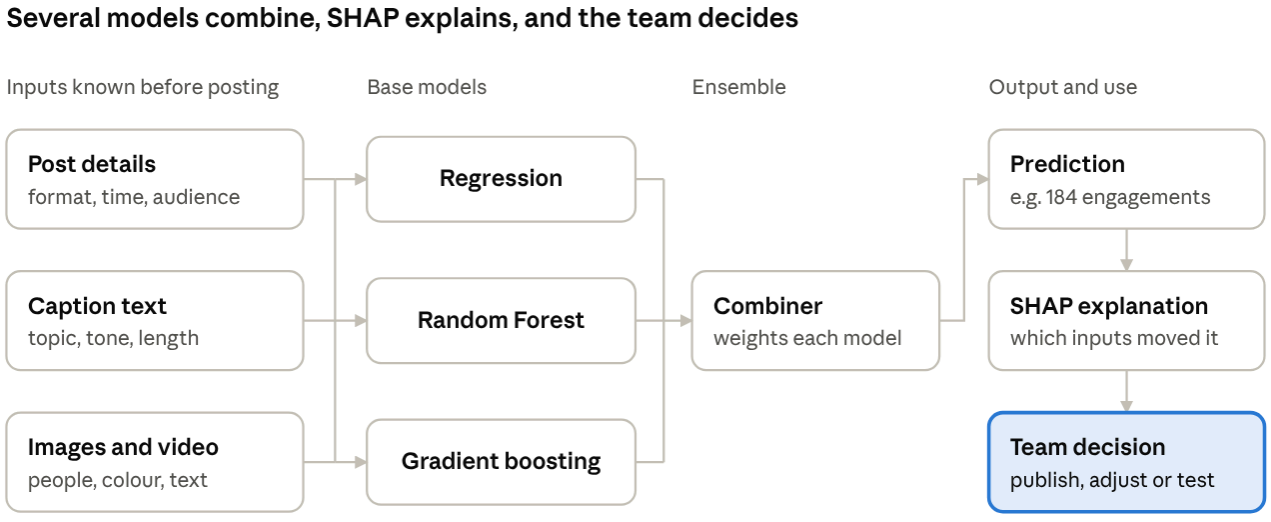
Ensembles solve this by combining many models. A **Random Forest** (Breiman, 2001) builds hundreds of trees, each on a slightly different random sample of the data, and averages their predictions; the averaging cancels out much of the instability of individual trees. **Gradient boosting** (Friedman, 2001) builds trees one after another, with each new tree concentrating on the errors left by the previous ones. XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017) are fast, widely used implementations of gradient boosting, and they appear repeatedly in the engagement-prediction studies reviewed in

Section 4. Support Vector Machines, which find the best boundary or curve separating or fitting the data, are another common alternative; the Chilean television study found them best on Instagram while XGBoost was best on Facebook.

7.2 Hybrid frameworks

A **hybrid** or stacked framework goes one step further by combining *different kinds* of models. For example, a Random Forest, an XGBoost model and a neural network might each make a prediction, and a final, simpler model learns how much weight to give each of them. A related idea is **soft voting**, in which each model produces a probability (for example, that a post will be "high engagement") and the final prediction is a weighted average of those probabilities. Hybrid frameworks are often paired with multimodal inputs: text embeddings, image features and structured metadata are each processed in the most suitable way and then fused.

The argument for hybrids is that different algorithms make different mistakes, so combining them tends to reduce error and improve generalisation. The argument against them, for most organisations, is cost and opacity: they need more data, more expertise and more maintenance, and their gains over a well-tuned single gradient-boosting model can be small. A team should adopt a hybrid only if it beats simpler models on genuinely unseen data by a margin that matters for decisions.



hybrid prediction framework · inputs, base models, combiner, explanation, human decision

Every input feeds every base model; a combiner weights their predictions, SHAP shows which inputs moved the result, and the final call stays with the communications team.

7.3 Context-aware and sequential models

Some engagement depends less on the content of a post than on the situation of the audience. The study by Peters and colleagues (2024) is the clearest recent demonstration. Using Long Short-Term Memory (LSTM) networks, a type of neural network designed to learn from sequences of events over time (Hochreiter and Schmidhuber, 1997), the authors modelled whether Snapchat users would engage actively (creating and sending content) or passively (viewing content). Adding momentary context, including smartphone connectivity status, location type, time and weather, raised the share of variation explained from 0.345 to 0.522. The authors interpret this as evidence of habit-driven behaviour that is triggered by context, and they report that connectivity status (whether a person is on mobile data or Wi-Fi) was among the most informative context features, plausibly because it signals whether someone is out and about, with moments worth sharing, or at home, scrolling.

Two further findings are relevant to practice. First, the authors' preprint abstract reports that a large share of predictive power could be achieved with very short behavioural histories once context was included (R^2 of about 0.44), which suggests that context-aware approaches can reduce the need to store long, privacy-sensitive records of individual behaviour. Second, this kind of modelling requires platform-level access to individual users' device data. It is directly relevant to how platforms rank and recommend content, and therefore to *why* the same post may reach different people at different times, but an individual organisation cannot replicate it with its own page analytics. For organisational teams, the transferable lesson is to treat audience context (time of day, connectivity costs, seasonal and event calendars) as legitimate predictors.

7.4 Why explainability matters

A prediction such as "expected engagement: 184" is of limited use on its own. A communications professional needs to know *why* the model expects that number, both to judge whether to trust it and to learn something about the audience. **Explainable AI** (XAI) refers to methods that make the reasoning of complex models visible.

The most widely used method is **SHAP** (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), which assigns each feature an importance value for a particular prediction. It borrows an idea from cooperative game theory, the Shapley value, which fairly divides a team's winnings among players according to their contribution. In this setting the "winnings" are the difference between the model's prediction for a specific post and its average prediction across all posts, and the "players" are the features. Each feature receives a positive or negative share showing how much it pushed this particular prediction up or down.

SHAP works at two levels. A **local explanation** shows why one post received its prediction: for example, "+22 because the format is a carousel, +15 because the topic is a funding call, −9 because it is scheduled for Saturday evening." A **global explanation** summarises, across all posts, which features matter most and in which direction.

7.5 How SHAP reveals non-linear relationships

A simple correlation coefficient can only say whether, overall, higher values of a feature go with higher or lower engagement. SHAP plots can show more complex shapes. If SHAP values for follower count rise steeply at first and then flatten, the model has learned diminishing returns. If a feature pushes predictions up for some posts and down for others with the same feature value, the model has learned an interaction: the feature's effect depends on something else, such as platform or topic. SHAP can also show that features which are individually weak, such as particular text or image dimensions, become influential in combination.

7.6 What explanations do not show

SHAP explains the *model*, not the *world*. If format is an important feature, that means format helped the model reproduce historical patterns. It does not prove that changing a post's format will cause engagement to change. When features are strongly correlated with one another (for example, "video" and "event content" if most videos are event highlights), SHAP may split credit between them in ways that are hard to interpret. Explanations should therefore be treated as prompts for investigation and, where possible, for controlled testing, not as proof of what works.

7.7 Context and habit, in the researchers' words

> Their findings are consistent with "context-contingent, habit-driven patterns of active and passive use." — Peters, Liu, Barbieri, Baten, Matz and Bos (2024)

For communications teams this is a useful reframing. Audiences do not evaluate each post afresh in a vacuum. They scroll, share and comment out of habits shaped by where they are, what device and connection they are using, and what time of day it is. A post that matches the audience's habitual context has an advantage that has nothing to do with its message.

7.8 Reading a SHAP explanation: an illustration

Suppose a model's average prediction across all past posts is 60 engagements, and for a planned post it predicts 92. A SHAP breakdown might look like this. The numbers are illustrative, not from a real model.

Feature of the planned post	Contribution to prediction
Starting point: average prediction	60
Format is a carousel	+14
Topic is a call for applications	+18
Contains a direct question	+5
Scheduled for Saturday evening	−9
Caption is unusually long	−4
Account has grown since last month	+8
Predicted engagement	**92**

The contributions add up exactly to the prediction, which is one of the mathematical properties that makes SHAP attractive. A communications officer reading this table can see at once that the topic and format are doing most of the work, and that moving the post to a weekday might be worth considering. What the table cannot say is whether moving the post would actually *cause* a gain of nine engagements; that would need testing.

7.9 Questions to consider

1. Would your team prefer a slightly less accurate model it can fully understand, or a more accurate one that needs an explanation tool?
2. What context about your audience (connectivity costs, working hours, religious or national calendars) might be as important as the content itself?
3. If an explanation tool said a feature mattered, how would you test whether changing it actually makes a difference?
4. Who in your organisation would be responsible for questioning a model's explanations when they seem implausible?

8. Evaluating a model: baselines and error measures

The right question about a predictive model is not "is it accurate?" but "is it meaningfully better than a simple alternative, on data it has not seen, by a margin that changes decisions?" Answering that requires a baseline and a small set of error measures.

8.1 The main error measures

Mean Absolute Error (MAE) is the average size of the prediction errors, ignoring whether they are too high or too low:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} \left| y_i - \hat{y}_i \right|$$

Here y_i is the actual engagement of post i, $\hat{y}_i$ is the predicted engagement, and n is the number of posts. An MAE of 13 means the model is, on average, 13 engagements away from the truth. MAE is easy to explain to colleagues and is expressed in the same units as the outcome.

Root Mean Squared Error (RMSE) squares each error before averaging and then takes the square root:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \left(y_i - \hat{y}_i \right)^2}$$

Because errors are squared, RMSE punishes large misses much more than small ones. It is useful when a badly wrong forecast is especially costly, for example when deciding whether to put paid promotion behind a post.

R-squared (R^2) compares the model's errors with the errors you would make by always predicting the average of the actual values:

$$R^2 = 1 - \frac{\sum_i \left(y_i - \hat{y}_i \right)^2}{\sum_i \left(y_i - \bar{y} \right)^2}$$

An R^2 of 0.88 means the model accounts for about 88 per cent of the variation in the outcome within the evaluated data. It should never be described as "88 per cent accuracy". R^2 can also be negative, which happens when a model does worse than simply predicting the mean.

Mean Absolute Percentage Error (MAPE) expresses each error as a percentage of the actual value and averages them:

$$MAPE = \frac{100}{n}\sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MAPE is intuitive and is the measure used by Moro et al. and Vairetti et al. However, it becomes unstable or undefined when actual values are zero or very small, which is common for comments and shares on smaller accounts. The choice of measure should follow the characteristics of the data and the decision being supported.

8.2 A worked example

Assume four test posts, A to D, received 80, 120, 100 and 160 engagements. A simple baseline predicts the account's historical average, 100, for every post. A regression model predicts 90, 110, 105 and 145.

Post	Actual	Baseline prediction	Baseline error	Model prediction	Model error
A	80	100	20	90	10
B	120	100	20	110	10
C	100	100	0	105	5
D	160	100	60	145	15

The resulting measures are:

Measure	Baseline	Regression model
MAE (engagements)	25.0	10.0
RMSE (engagements)	33.2	10.6
MAPE	19.8%	8.8%
R^2	−0.26	0.87

The model reduces MAE from 25 to 10, a 60 per cent reduction ((25 − 10) ÷ 25 × 100). Its RMSE improvement is even larger, because the baseline badly misses post D. The baseline's R^2 is negative because a flat prediction of 100 is worse than the true average of these four posts (115). These numbers are illustrative only; they are not results from an experiment, and four posts are far too few for real evaluation.

8.3 Why the baseline is essential

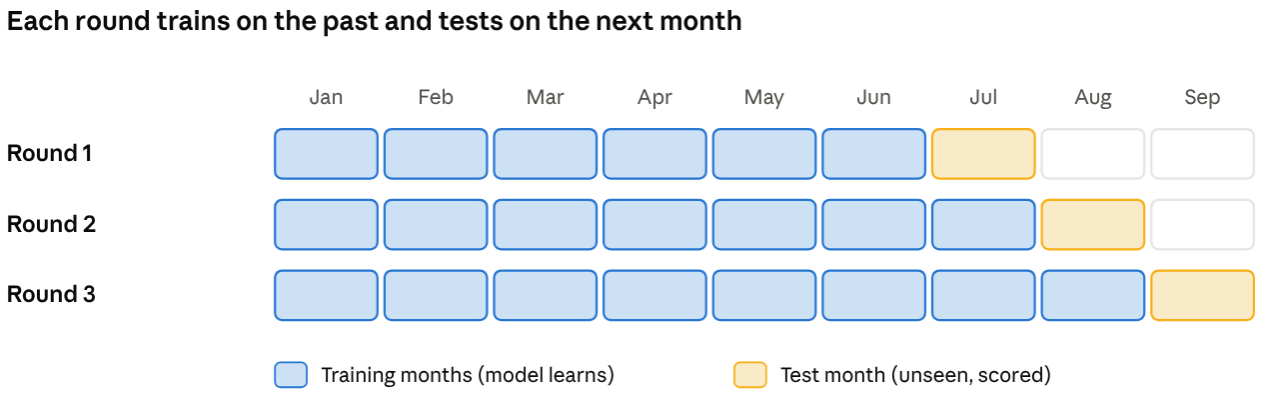
A machine-learning model can produce impressive-looking statistics while adding little practical value. If the account's historical average, or its average for the same format over the previous month, predicts nearly as well as the model, the extra complexity is not justified. The Fragile Families challenge described in Section 4.5 is a sobering example of sophisticated methods barely beating a simple benchmark. Useful baselines include the historical mean, the historical median (more robust to viral outliers), and a rolling average of the last 10 or 20 posts.

The leaking cup. Testing on unseen posts is filling the cup with water before serving anyone from it, and holding it up to the light. The baseline is the plain cup the team already owns. A new model earns its place only if it holds noticeably more than that plain cup, on

water it has never touched.

8.4 Testing in time order

Randomly splitting posts into training and test sets is common, but for social media forecasting it can be misleading. If a random split puts some March posts into training and other March posts into testing, the model gets an unrealistic glimpse of the conditions it is meant to predict. A **chronological split**, training on older posts and testing on newer ones, better imitates the real task. A stronger version, sometimes called rolling-origin or time-series cross-validation, repeats this several times: train on January to June and test on July; train on January to July and test on August; and so on. The spread of results across these repeats shows how stable the model is over time.



rolling-origin evaluation · three rounds, each tested on the following month

Each round imitates a real forecast: the model sees only the months before the test month, and the spread of scores across rounds shows whether its accuracy holds up over time.

8.5 A warning from the largest prediction challenge in social science

> The best predictions "were only slightly better than those from a simple benchmark model." — Salganik et al. (2020)

This finding came from 160 teams who built models for six life outcomes using a rich, high-quality dataset. The authors also found that prediction error depended much more on *which family* was being predicted than on *which technique* was used. The lesson transfers directly to social media: the choice of algorithm usually matters less than the quality of the data, the definition of the outcome, and honest comparison with a simple benchmark.

8.6 Interpreting results in plain language

When reporting a model to colleagues, translate the numbers into statements people can act on:

- "On last quarter's posts, which the model had never seen, its forecasts were off by about 14 engagements on average, compared with 22 for our usual-median rule."
- "It does well for routine announcements but under-predicts event videos, so treat its video forecasts with caution."
- "It explains roughly half of the variation between posts; the rest depends on things it cannot see."

These are the kinds of sentences that let non-specialists decide how much weight a forecast deserves. (The figures here are illustrative.)

8.7 Questions to consider

1. What is the simplest forecast your team could make today, and how accurate would it have been last quarter?
2. Which error matters more for your decisions: being slightly wrong often, or very wrong occasionally?
3. Do any of your outcomes (such as shares on a small account) often equal zero, making percentage errors misleading?
4. If a model only beat the baseline by a small margin, would that margin be worth the effort of maintaining it?

9. A proposed empirical design for a communications team

This section translates the preceding ideas into a study that a communications team could carry out on its own authorised data. It is a research design, not a report of results: no model has been trained or validated here, and that distinction should remain explicit in any internal reporting.

9.1 Research question and hypotheses

> Can pre-publication characteristics of an organisation's social media posts predict engagement within 48 hours more accurately than a historical-average baseline?

Three testable hypotheses follow:

1. **H1.** A linear regression model using pre-publication features will achieve a lower MAE on a chronologically held-out test set than the historical-median baseline.

2. **H2.** A gradient-boosting model will achieve a lower MAE than the linear regression model on the same test set.
3. **H3.** Adding simple text features (topic category, call to action, caption length) will reduce MAE beyond structured metadata alone.

Each hypothesis can be rejected, which is what makes the study informative. If H2 is rejected, for example, the team learns that the simpler, more interpretable model is good enough.

9.2 The dataset

Each row represents one published post. A suggested structure:

Column	Example	Known before publication?
Post ID	FB-2025-0412	Yes (identifier only, not a predictor)
Platform	Facebook, Instagram, LinkedIn, X	Yes
Account	Main page, programme page	Yes
Publication date and time	2025-04-12 10:30	Yes
Content format	Image, carousel, Reel, video, text, link	Yes
Topic or campaign category	Grant call, partner story, event, policy	Yes
Caption length (characters)	412	Yes
Hashtag and mention counts	3; 2	Yes
Call to action present	Yes / No	Yes
Paid or organic	Organic	Yes
Audience size at time of posting	18,450 followers	Yes
Account's median engagement, previous 20 posts	63	Yes, if computed only from earlier posts
Engagements within 48 hours	87	No (this is the outcome)
Lifetime reach and impressions	9,800; 14,200	No (never use as a predictor)

Data preparation should include removing duplicates, standardising platform and format labels, checking time zones, investigating missing and impossible values, inspecting extreme outliers rather than deleting them automatically, and documenting every transformation so the analysis can be repeated.

The leaking cup. Building this dataset is, in itself, the first repair of the knowledge leak. In many organisations the most valuable lessons about communication live in people's memories, scattered inboxes and the write-ups of those who attended an activity, and they drain away as staff move on. A single, consistent record of every post, what was planned and what happened is how a team stops pouring its experience onto the table.

9.3 How much data is enough?

There is no fixed rule, but small datasets limit what can be learned. A useful heuristic for regression is to have at least 10 to 20 posts for every predictor included. A team with 500 posts can therefore support a modest model of perhaps 15 to 25 predictors; it cannot reliably support thousands of text-embedding dimensions. Where posts are spread across several platforms, it is usually better to start with the single platform that has the most history.

9.4 Procedure

1. Collect post-level data from authorised organisational accounts for a defined period, for example January 2024 to June 2026.
2. Define the outcome as total engagements (reactions, comments, shares and saves) within 48 hours of publication, and record it consistently.
3. Create predictor variables using only information available before publication, applying the leakage test in Section 5.2 to each one.
4. Split the data chronologically: roughly the earliest 80 per cent for training and the most recent 20 per cent for testing. With 500 posts, that is about 400 training and 100 test posts.
5. Calculate the baselines (historical mean, historical median and rolling average of the last 20 posts) for every test post.
6. Fit a linear regression model on the training data, using a logarithm of engagement if the distribution is highly skewed.
7. Fit a Random Forest and a gradient-boosting model, tuning their settings with time-ordered cross-validation inside the training data only.
8. Generate predictions for the test posts and compare all models and baselines using MAE, RMSE and R^2 (and MAPE where actual values are rarely near zero).
9. Analyse errors by platform, format, topic, account size and month, to see where the model works and where it fails.
10. Use SHAP to produce global and local explanations for the best model, and review them with the communications team for plausibility.

9.5 Pre-publication and post-publication prediction

The design above answers a planning question: *how is this post likely to perform before we publish it?* A second, different model could answer a monitoring question: *after the first hour, how much engagement is this post likely to have after 48 hours?* That model could legitimately use early signals such as first-hour reach, early comments and shares, and early video retention. It will usually be considerably more accurate, because early

performance is strongly linked to later performance. But it supports different decisions, such as whether to boost a post, reply actively to comments, or share it across partner channels. Any dashboard or report should state clearly which kind of prediction it shows.

9.6 A results template

When the study is complete, its main findings fit in one table. Filling it in honestly, including when the more complex model does not win, is the core of the study.

Model	MAE (test set)	RMSE (test set)	R^2 (test set)	Beats median baseline?
Historical median (baseline)				—
Rolling average, last 20 posts				
Linear regression				
Random Forest				
Gradient boosting				
Gradient boosting + text features				

9.7 Common mistakes to avoid

- **Testing on the training data.** Every number in the results table must come from posts the model never saw.
- **Random rather than time-ordered splits.** They flatter the model by mixing future conditions into training.
- **Tuning on the test set.** Adjusting the model repeatedly until the test score looks good quietly turns the test set into training data.
- **Including viral outliers without thought.** One extraordinary post can dominate a small dataset. Investigate it, consider a log scale, and report results with and without it.
- **Changing definitions midway.** If "engagement" was defined one way in 2024 and another in 2025, the model learns the change in definition, not the audience.
- **Reporting only the best model.** Showing all models, including those that failed, is what makes the conclusion credible.

9.8 A minimal starter toolkit

Need	Low-cost option
Data export	Native analytics exports from each platform (Meta Business Suite, LinkedIn page analytics, and similar)
Data cleaning	A shared spreadsheet with fixed column definitions
First regression	Spreadsheet regression tools or Google Colab with Python
Machine-learning models	The free scikit-learn library in Python
Explanations	The open-source SHAP library
Documentation	A one-page model card (see Section 10.5)

9.9 Questions to consider

1. How many posts per platform do you have with consistent data, and for what period?
2. Who in your team or organisation has the skills to run the analysis, and who could review it critically?
3. Which of the three hypotheses would surprise you most if it turned out to be false?
4. How will you share the results, including any disappointing ones, with colleagues and leadership?

10. Limits, risks and responsible use

A technically accurate engagement model can still lead to poor communications. This section sets out the main limits and the safeguards that responsible use requires.

10.1 Engagement is not effectiveness

Engagement is one possible outcome of communication, and often not the most important one. A post that provokes intense reactions may do nothing to improve public understanding, build trust, increase programme participation, support fundraising, strengthen institutional credibility or advance an organisation's mission. Conversely, a carefully argued policy brief may receive modest engagement while reaching exactly the decision-makers it was written for.

The risk is captured by Goodhart's law, often paraphrased as: when a measure becomes a target, it ceases to be a good measure. An organisation that optimises relentlessly for predicted engagement may drift towards sensational, emotionally provocative or repetitive content, and away from the harder, slower communication its mission requires. The communications objective should therefore choose the metric, not the other way round:

Communications objective	More suitable outcome to predict
Awareness	Reach or unique viewers
Conversation and community	Comments and replies
Dissemination through networks	Shares and reposts
Driving action (applications, downloads)	Link clicks and conversions
Long-form storytelling	Watch time and completion rate
Credibility with specific audiences	Engagement from target groups (where measurable), qualitative feedback

Over time, organisations may build separate, small models for separate objectives rather than reducing all success to one engagement score.

> "When a measure becomes a target, it ceases to be a good measure." — Marilyn Strathern (1997), summarising Goodhart's law

Strathern was writing about university examinations, but the warning fits social media closely. Engagement is a reasonable signal of attention while nobody is chasing it. Once a team is rewarded for engagement alone, it becomes tempting to post more provocative, more emotional or more frequent content, and the number rises while the communication it was meant to reflect may get worse. A predictive model can accelerate this drift, because it makes "what scores well" visible and easy to optimise.

Example. A model shows that posts featuring celebrities at events attract high engagement. A team that follows the model blindly might shift its content towards celebrity photographs and away from the community voices and programme results that its mission depends on. The model would be right about engagement and wrong about what matters.

The leaking cup. A cup that overflows at the rim looks generous, but water running down the outside nourishes no one. High engagement can be exactly that: reactions spilling over while the people the communication was meant for never drink. Chasing the score

makes it worse, because the team starts pouring faster to make the overflow look impressive. This is a leak of impact disguised as success.

10.2 Correlation is not causation

As explained in Section 2.8, a predictive model uses historical correlations. If the model says carousels are associated with higher engagement, it may be because carousels genuinely work better for this audience, or because the team happened to use carousels for its most interesting stories. The only reliable way to learn whether a change *causes* better results is to test it deliberately: for example, by alternating formats for comparable content over several weeks, or by using platform A/B testing tools where they exist. Predictive models are excellent at generating hypotheses for such tests; they are not a substitute for them.

A simple test design. Suppose a model suggests that carousels outperform single images for programme stories. Over eight weeks, the team could alternate formats for comparable programme stories, one week carousel, one week single image, keeping topic type, posting day and time as similar as possible. Comparing the two groups at the end gives much stronger evidence than the original correlation, because the team, not history, decided which posts got which format. With small numbers the result will still be uncertain, but repeated over time such tests build genuine, causal knowledge about the audience.

10.3 Drift: the world changes

Platforms regularly change the algorithms that decide what people see. Audiences grow, age and shift between platforms. Organisational priorities and visual styles change. All of this means that a model trained on 2024 data may quietly become less accurate in 2026. This is called **model drift**. The safeguard is routine: compare predictions with actual outcomes every month or quarter, retrain the model on recent data, and retire it if it no longer beats the baseline.

Warning signs that a model may be drifting include:

- forecast errors that grow steadily over several months;
- errors that are consistently in one direction, for example the model always over-predicts;
- a platform announcing changes to how feeds rank content;
- a sudden change in audience size, for example after a campaign or an account merger;
- a change in the team's own content strategy, such as a new visual identity or posting schedule.

The leaking cup. Drift is a crack that widens slowly. A model that was watertight last year begins to seep as platforms, audiences and strategies change, and because the change is gradual, nobody notices until the puddle is large. Regular checking is the habit of running a finger around the rim.

10.4 Privacy and data minimisation

Post-level analytics from an organisation's own accounts are aggregate and usually low-risk, but projects can creep towards collecting data about individual commenters or followers. Only authorised data should be used, in line with platform terms of service. Personal identifiers that are not needed for prediction should be removed or anonymised. The existence of a variable does not justify collecting it. In Uganda, the Data Protection and Privacy Act, 2019 sets legal obligations for collecting and processing personal data, and comparable laws exist in Kenya, South Africa, Nigeria and elsewhere. The NIST AI Risk Management Framework, released in January 2023, offers a widely used general structure for identifying and managing such risks across the life of an AI system.

A useful discipline is to ask, for every column in the dataset: *Do we need this to answer our research question?* Aggregate, post-level figures (format, topic, total engagements) almost never identify individuals. Lists of commenters' names, profile details or private messages almost always do, and are rarely needed for engagement prediction. Leaving them out reduces legal risk and protects the trust of the communities an organisation serves.

10.5 Transparency and human judgement

Everyone who uses a model's outputs should know what it predicts, what information it uses, how it was tested, where it performs poorly and when it was last updated. A one-page "model card" kept alongside the dashboard is sufficient for most teams.

Human professionals understand context that does not exist in historical data: political developments, emergencies, bereavements, sensitive anniversaries, stakeholder relationships, cultural meaning, ethical concerns and programme strategy. A model trained on past posts cannot know that a scheduled celebratory post should be postponed because of a national tragedy that morning. Predictions should therefore always be one input into a human decision, never the decision itself. Strategically important content, such as a statement on a matter of principle or a report a partner is depending on, may need to be published regardless of its predicted engagement.

The leaking cup. A model can point to where the cup is likely to leak. It cannot hold the cup. It is people who decide what to pour, when, and for whom, and who know when a post must be served regardless of how much might spill, because the person waiting for it matters more than the measurement.

What the researchers themselves say

The studies reviewed in this paper repeatedly frame prediction as an aid to people, not a replacement for them. Their own words make the point.

> Understanding why a model makes a prediction "can be as crucial as the prediction's accuracy." — Lundberg and Lee (2017), the creators of SHAP

This is an argument for transparency. A forecast that nobody can interrogate cannot be weighed against the things the model does not know.

> Their Facebook model "may be used to support manager's decisions on whether to publish a post." — Moro, Rita and Vala (2016)

Note the verb: *support*. Even in one of the earliest engagement-prediction studies, the authors positioned the model as an input to a manager's choice.

> Posting "without a strategy in place would not lead to higher awareness." — Bhattacharjee, Suresh and Maiti (2026)

Prediction cannot substitute for strategy. A model can say which post is likely to attract reactions; only the organisation can say which reactions it actually wants.

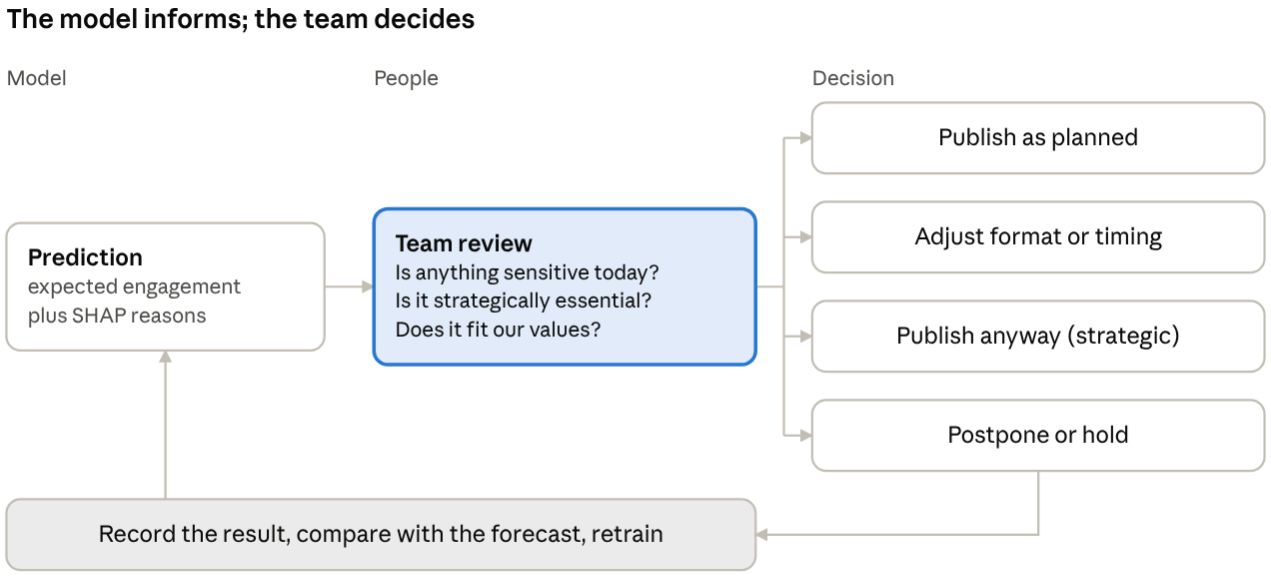
> "Test data should never be used to make choices about the model." — scikit-learn documentation

Transparency includes honesty about how a model was tested. A team that cannot say how its model was evaluated cannot say how far to trust it.

> The NIST framework exists "to incorporate trustworthiness considerations" into how AI is designed, used and evaluated. — NIST AI Risk Management Framework

Trustworthiness here is not an abstract ideal. In a communications team it means knowing a tool's limits, documenting them, and keeping a person accountable for every public statement.

Where the human decision sits



human-in-the-loop decision flow · prediction, team review, four outcomes, learning loop

The prediction and its explanation flow into a team review, which can lead to four different outcomes; whatever is chosen, the result is recorded so the next forecast is better informed.

The diagram shows that a low prediction does not lead automatically to "don't publish". It leads to a conversation. Sometimes the team adjusts the format or timing. Sometimes it publishes anyway, because the content is strategically essential. Sometimes it holds the post because of events the model could not have known about. In every case, recording what happened turns each decision into new evidence.

A model card: what to document

A "model card" is a short, plain-language record kept beside any predictive tool. For a communications team it can be a single table:

Field	What to record	Example
Purpose	The decision the model supports	Planning Facebook posts a week ahead
Outcome predicted	Exact metric and time window	Total engagements within 48 hours
Inputs used	Every predictor, and confirmation each is known before posting	Format, topic, day, hour, caption length, follower count

Field	What to record	Example
Data and period	Source, date range, number of posts	Page export, Jan 2024–Jun 2026, 512 posts
How it was tested	Split method, baseline and results	Chronological 80/20; MAE 18 vs baseline 27
Known weaknesses	Where errors are largest	Under-predicts event videos; weak for new topics
Owner and review date	Who maintains it and when it is next checked	Communications lead; quarterly

The example values are illustrative. The point is that anyone in the team, or a manager or partner, can see in one glance what the tool does and does not do.

An illustrative scenario

Imagine a model predicts that a post announcing a partner's research report will receive below-average engagement: it is a link post, published on a Friday, about a policy topic that has historically drawn few reactions. A purely automated system might deprioritise it. A communications officer reviewing the forecast knows things the model does not: the report was requested by a government ministry, the partner is relying on the announcement, and a handful of policymakers, not thousands of followers, are the real audience. The officer publishes as planned, but uses the SHAP explanation to make small changes, such as converting the link post into a carousel of key findings and moving it to a Tuesday. Afterwards, the result is recorded. If the adjusted post outperforms the forecast, the team has learned something; if it does not, the team has still served its partner. Either way, the decision was a human one, informed by evidence.

Questions to ask yourself

These questions can guide a team, or an individual reader, before trusting or acting on any engagement prediction:

1. What exactly is this model predicting, and is that the outcome my communication is actually meant to achieve?
2. Would every input the model used have been known at the moment I needed the forecast?
3. Does the model beat a simple average on posts it has never seen, and by how much?
4. Where does the model perform worst, and is my current post one of those cases?
5. What does the explanation say pushed this prediction up or down, and does that make sense to someone who knows our audience?

6. Is there anything happening today, in the news, in our communities or with our partners, that the model cannot know about?
7. Is this content strategically important enough to publish regardless of the forecast?
8. Would I be comfortable explaining to a partner or a community member why we made this decision?
9. Am I starting to choose content because it will score well, rather than because it matters?
10. When did someone last check whether this model is still accurate?

10.6 Specific considerations for African and civil-society contexts

Several features of the communications environment in East Africa and in civil-society organisations more broadly deserve attention.

- **Small datasets.** Many organisations publish a few posts a week per platform. Two years of history may yield only a few hundred observations, which favours simple, interpretable models and categorical outcomes over complex deep learning.
- **Measurement gaps.** Platform restrictions, VPN use and shared devices can make audience data noisy, as the Ugandan Facebook example in Section 3.1 shows.
- **Language.** Content often mixes English with local languages. Off-the-shelf sentiment and topic tools may misread it, so text features need local testing.
- **Connectivity and cost.** Mobile data costs shape when and how audiences engage, especially with video. The context-aware evidence in Section 7.3 suggests these factors are not minor; they may be among the strongest drivers of behaviour.
- **Dignity and representation.** Civil-society communication often tells the stories of communities and individuals. A model may learn that certain kinds of imagery attract high engagement. Whether such imagery is respectful and consistent with the organisation's values is a judgement that no engagement model can make.
- **Data from partners.** Content produced by partners, grantees or event participants is part of the record but may follow different patterns. Where such content is analysed, permissions and attribution should be clear.

10.7 Questions to consider

1. If your team's performance were judged only on engagement, what kinds of content would you stop producing?
2. Which of your current "what works" beliefs could be tested with a simple, deliberate experiment over the next two months?
3. When did your main platforms last change how they rank content, and would your historical data still reflect current conditions?

4. Does your dataset contain any information about individual people that you do not actually need?
5. Which local laws and organisational policies govern how you collect and use social media data?
6. In your context, which audiences are underrepresented in platform data, and how might that skew what a model learns?
7. Who should be involved, beyond the communications team, in deciding how predictions are used?

11. Implications for communications practice

The evidence points to a clear practical role for predictive modelling: a decision-support tool that helps teams test assumptions, plan more systematically and learn from their own history, while editorial judgement stays with people. The following ten steps turn the research framework into a workflow.

1. **Define a specific prediction problem.** Do not begin with "we want AI to improve our social media". Begin with a measurable question, such as "can we predict the number of engagements a Facebook post will receive within 48 hours, before we publish it?" Specify the outcome, platform, time window and unit of analysis.
2. **Assemble a suitable dataset.** Export authorised, post-level historical data, one row per post. Start with the platform that has the longest and most consistent history.
3. **Prepare and inspect the data.** Remove duplicates, standardise categories, check dates and time zones, investigate missing values and outliers, create the fixed performance window, and record every change you make.
4. **Keep only information available before publication.** Apply the leakage test to every column. Reach, impressions and lifetime metrics belong in the outcome, never in the predictors of a pre-publication model.
5. **Build a simple baseline.** Calculate the historical median engagement and a rolling average of recent posts. Any model must beat these to be worth using.
6. **Train a first regression model.** This can be done in Excel, Google Sheets, Google Colab or Python with scikit-learn. A transparent model the team understands is worth more than a sophisticated one nobody can question.
7. **Evaluate on unseen, more recent posts.** Use a chronological test set and report MAE, RMSE and R^2 alongside the baseline.
8. **Investigate the errors.** Ask where the model fails: on video, on small accounts, on event posts, during unusual periods, on particular topics. Error analysis is often more instructive than the headline score.

9. **Translate predictions into questions, not verdicts.** If the model expects a post to underperform, ask why, and consider format, timing, topic and audience fit. A low prediction should prompt reflection, not automatic rejection of the content.
10. **Review and retrain regularly.** Compare predictions with outcomes each month or quarter, add new data, and check for drift. Retire the model if it stops beating the baseline.

11.1 What a team can realistically gain

Even a modest model can change practice in useful ways. It forces the team to define success explicitly. It replaces anecdotes ("videos always do well") with tested patterns. It creates a disciplined record of what was planned and what happened. And through explanation tools, it can surface patterns that experienced staff had sensed but never articulated, or challenge beliefs that the data do not support.

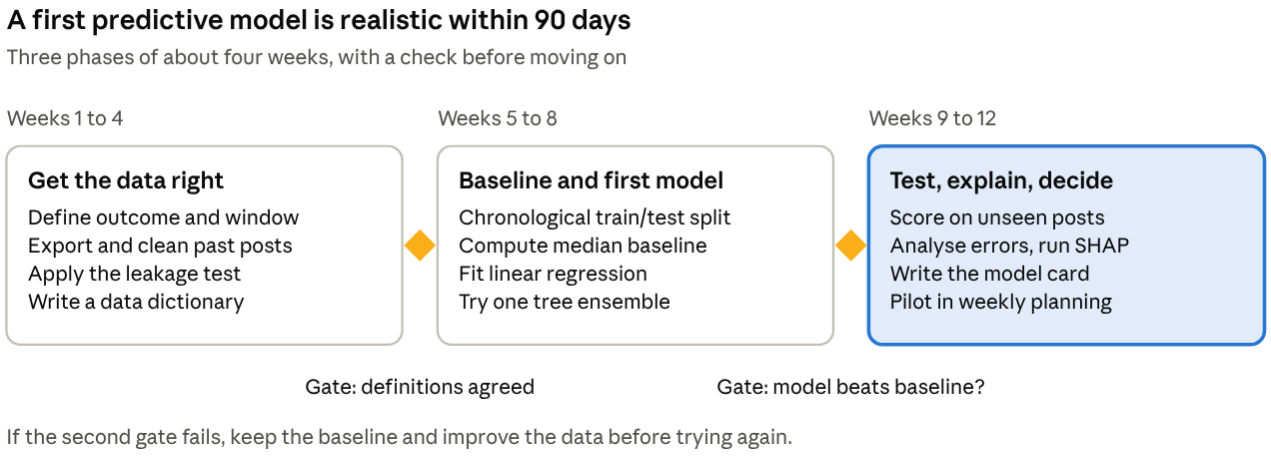
The realistic expectation, based on the studies reviewed here, is a forecast that is wrong by perhaps 15 to 30 per cent on average for routine content, that clearly beats a naive average, and that is weakest precisely for the rare posts that perform exceptionally well or badly. That is good enough to improve scheduling, format choices and resource allocation. It is not good enough to justify letting a model decide what an organisation says.

11.2 Where predictive work connects to broader communications systems

Prediction depends on consistent, well-structured records. Organisations that already maintain a content calendar, tag posts by programme or campaign, and collect structured write-ups of their activities are well placed, because these records supply the topic and category variables that models need. In that sense, the most valuable first investment is often not in AI at all, but in disciplined data practices: consistent labels, fixed reporting windows, and a shared spreadsheet that records each post's planned features alongside its results.

The leaking cup. The two leaks are connected. When knowledge leaks, because results are not recorded and lessons not shared, the same impact leaks recur year after year, because nobody remembers where they were. Capturing knowledge is therefore not separate from improving impact; it is how impact improves. A team that records each pouring carefully slowly learns the shape of its own cup.

11.3 A realistic first 90 days



a 90-day roadmap · three phases with two decision gates

The roadmap is deliberately modest: most of the value in the first three months comes from getting definitions and data right, and the most important moment is the second gate, where the team decides honestly whether the model has earned a place in planning.

11.4 What success looks like after a year

A team that follows this approach for a year should not expect a machine that tells it what to post. It can reasonably expect:

- a clean, consistently defined record of every post and its results;
- a tested understanding of which formats, topics and timings matter for its own audience;
- a simple model that beats the baseline for routine content, with known weaknesses;
- a handful of deliberate experiments that have confirmed or overturned long-held assumptions;
- reporting that distinguishes clearly between what happened, what was predicted, and what was caused.

11.5 Questions to consider

1. Which single planning decision would benefit most from a reliable forecast in your team?
2. What is the smallest version of this project you could complete in the next three months?
3. Who would own the model, and who would challenge it?
4. How will you know, a year from now, whether the effort improved your communication, and not just your engagement figures?

12. Conclusion

Machine learning offers communications professionals a way to move from purely retrospective analytics towards structured, testable forecasting. The evidence reviewed in this paper supports three conclusions.

First, social media engagement is predictable to a useful but limited degree. Studies of organisational Facebook and Instagram accounts report average forecast errors of roughly 15 to 30 per cent using information available before publication, and large-scale platform research shows that context can substantially improve predictions of user behaviour. At the same time, research on social influence and on the limits of prediction shows that exceptional successes remain hard to foresee.

Second, the credibility of any prediction depends far more on research design than on algorithmic sophistication. A fixed and meaningful outcome, predictors that genuinely exist at the moment of prediction, chronological testing on unseen data, and honest comparison with a simple baseline matter more than whether the model is a regression, a gradient-boosting ensemble or a hybrid neural architecture. Natural language processing and computer vision can extend what is measurable, and explainable AI techniques such as SHAP can make complex models intelligible, but none of these rescues a study that leaks future information or skips the baseline.

Third, predictive performance must never be confused with communications effectiveness. Engagement is not the same as trust, understanding, influence or impact. Correlations learned from the past are not causal rules, and models drift as platforms and audiences change.

The strongest application of AI in communications is therefore not an automated system that decides what an organisation should say. It is a decision-support system that helps communications professionals ask better questions of their own history: which assumptions hold, which do not, and where attention and resources are best spent. Used in that way, with transparency, care for privacy and respect for context, predictive modelling can make communications practice more systematic without making it less human. The final judgement about what should be communicated, to whom, when and why remains a human responsibility.

The leaking cup. No communications team will ever own a cup that does not leak. Audiences are busy, platforms are opaque, and the most important moments are often the least predictable. The aim of this paper has been more modest and more practical: to stop measuring only the puddle, and to start learning where our own cup leaks, both the impact that never reaches people and the knowledge we never keep. Predictive modelling, used with honesty and care, is one way of holding more of what we pour. The rest depends, as it always has, on the hands that hold the cup.

12.1 Directions for further research

Several questions remain open and would particularly benefit from research in African and civil-society settings:

- **Small-data methods.** How well do engagement models perform for organisations with only a few hundred posts, and which simple methods work best at that scale?
- **Multilingual content.** How much predictive value do text features add when captions mix English with Luganda, Swahili or other local languages?
- **Beyond engagement.** Can models predict outcomes closer to impact, such as applications received, event attendance or policy uptake, using social media signals?
- **Context and cost.** How do mobile data costs, connectivity and device sharing shape when and how African audiences engage?
- **Causal evidence.** What do deliberate, repeated experiments reveal about the formats and timings that genuinely change audience behaviour?

12.2 Closing questions for the reader

1. After reading this paper, which of your organisation's assumptions about social media would you most like to test?
2. What would "good enough" accuracy look like for your planning decisions?
3. Which outcomes, other than engagement, best reflect the purpose of your communication?
4. How will you make sure that any model your team builds remains a tool for judgement, and never a substitute for it?

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